

Age of Information and Reliability Trade-offs

Conceptual Understanding

Age of Information (AoI) explanation:

- a. Age of Information to me means how fresh the information is that has been sent to the receiver. It's the time elapsed since the last received packet was generated at the source. This is important because old information does not help inform current mission critical operations. For example in the Industrial Internet of Things, decisions are usually automated and based on real-time sensor data. If a PLC controlling a robot arm receives stale data it could perform actions that might be incorrect or dangerous.

AoI-oriented vs deadline-oriented traffic:

- a. The main difference between the two is that AoI traffic focuses primarily on continuous freshness while Deadline-Oriented traffic focuses on timely delivery of specific events. For AoI a real world example could be in Environmental Monitoring particularly a sensor measuring the temperature of a chemical vat. The system needs the most recent reading to ensure that the right countermeasures are applied in the event something goes wrong. Emergency Shutdown Systems would be a good way to explain the importance of Deadline-Oriented traffic. Take a pressure sensor for example. If this sensor detects that a pipe is about to burst, it needs to send a "Shutdown" signal to the machinery. In this case its critical for the signal to arrive within a specific timeframe (5 ms for example) to trigger the safety valve. If it arrives even 1ms slower, it has missed its "deadline" which could have catastrophic effects.

Analysis and Insights

Influential Key Factors:

- a. The key factors that appear to influence the AoI-PLP trade-off based on the dataset are number of nodes and capture threshold. The nodes are the devices transmitting information and the capture threshold helps determine the capture

effect. In other words it determines the central controller's ability to successfully decode a packet if the received Signal-to-Interference-plus-Noise Ratio (SINR) exceeds the pre-determined capture threshold.

Proposed Strategies:

There are two strategies that can be used for optimizing network performance to balance data freshness and reliability.

- a. The first would be enhanced channel access strategies which refer to how devices compete for and use the wireless medium.
 - a. Priority-Based Access:
 - Higher priority is assigned to newer or time-sensitive packets.
 - Effect:
 - Aol drops as fresh packets get through faster.
 - Potential increase in PLP for low-priority traffic.
 - b. Adaptive Contention Window:
 - Adjust in real time how vigorously nodes compete for the channel.
 - In the case of high congestion, increasing the contention window would lead to fewer collisions and lower PLP.
 - In the case of high Aol, decreasing the contention window provides faster access which lowers Aol.
 - c. Scheduled/Hybrid Access:
 - Reserve time slots only for critical nodes. The rest use contention-based access.
 - The effect of this strategy is the potential lowering of both AOI and PLP.
- b. The second would be Queue Management Strategies. These control what happens to packets waiting to be transmitted.
 - a. Drop Policies:
 - This strategy implements a Last-Come-First-Serve (LCFS) framework with replacement. New packets replace older ones in the queue.
 - With this the system will always transmit the freshest data which will lower the Aol and older packets get dropped which increases PLP but could still be acceptable in many IoT cases.
 - b. Queue Prioritization:
 - Separates queues:
 - High-priority (fresh/critical data)

- Low-priority (less time-sensitive)
 - The scheduler serves high-priority first
- c. Preemption Strategies:
 - Ongoing transmission or queued packet can be interrupted by fresher data.
 - This can aggressively minimize AoI and may increase PLP for interrupted packets.

Potential Real-World Application:

- a. A real-world application using the insights could be a system that detects when the balance between PLP and AoI is off. It then alerts a person or a controller that can use one of the strategies i've mentioned in the previous section to bring the system back to its equilibrium. This would ensure that the system maintains a good balance between AoI and PLP when receiving information.

Comparison: DL & Random Forest Models

- a. For starters it seems like the deep learning model does ok with predicting PLP but not so well with AoI as per their R2 scores. The Random Forest model achieved an accuracy of around 70% which in my opinion is not bad considering the complexity of the features used to train it. Overall the deep learning model performs significantly better with more than 2 million less in MSE. Even though this MSE value is still exorbitantly large it is an improvement. Furthermore it improved the R2 score for AoI compared to the random forest model significantly as well. Even though both outcomes still show that the model is not performing well in predicting Age of Information, we are moving in the right direction. I tried many different layer architectures and hyper-parameters to try to improve the scores but unfortunately I had no luck. The larger I made the network and the higher the epochs or batch sizes the worse it performed.